

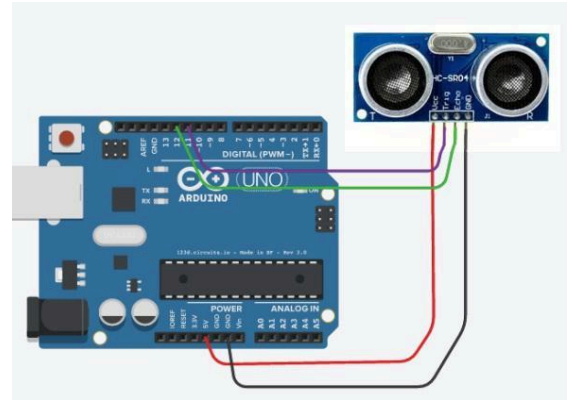
Experiment 3: Ranging and Error Characterization

Analyze Ultrasonic and Infrared Distance Data to Model Sensor Accuracy and Identify Technology-Specific Limitations

Aim: To measure distance using distance sensors and analyze the measured data using an Arduino.

Hardware:

- Arduino UNO
- HC-SR04 Ultrasonic Sensor
- Jumper wires/USB cable
- Measuring scale
- Computer with Arduino IDE



Wiring: (HC-SR04 sensor)

Sensor	Arduino
V _{CC}	5 Volts
GND	GND
Trig	D9
Echo	D10

Experimental Procedure:

1. Connect the sensor to Arduino.
2. Upload the program and open Serial Monitor.
3. Place an object at known distances.
4. Record distance obtained from the sensor.
5. Record IR analog value.
6. Repeat the experiment various times to compare readings and calculate error statistics.
7. If possible, repeat the experiment by replacing the target with various objects and on various surfaces (Soft/Inclined).

Theory:

The HC-SR04 ultrasonic sensor includes a transmitter & a receiver. This sensor is used to find out the distance from the objective. Here the amount of time taken to transmit and receive the waves will decide the distance between the sensor and an object. This sensor uses sound waves by using non-contact technology. The range of this sensor is available between 2 to 400 cms.

The basic working principle:

- The transmitter emits 40 kHz frequency signals. The echo pin goes high when the signal is sent and drops low when it's received.
- Receiver receives the reflected signal; the high-level duration of the echo pin represents the time from sending to receiving the ultrasonic pulse.

HC-SR04 Ultrasonic Sensor Pin Configuration:

This sensor includes four pins and the pin configuration of this sensor is discussed below.

- Pin1 (V_{cc}): This pin provides a +5V power supply to the sensor.
- Pin2 (Trigger): This is an input pin, used to initialize measurement by transmitting ultrasonic waves by keeping this pin high for 10us.
- Pin3 (Echo): This is an output pin, which goes high for a specific time period, and it will be equivalent to the duration of the time for the wave to return back to the sensor.
- Pin4 (Ground): This is a GND pin used to connect to the GND of the system.



Calculations:

$$Distance = \frac{Velocity \times Time}{2}$$

where

Velocity of sound(v) - 343 m/s

Mean Distance

$$\bar{x} = \frac{x_1 + x_2 + x_3 + \dots + x_n}{n}$$

Absolute Error

$$E = |Measured - Actual|$$

Percentage Error

$$\%Error = \frac{|Measured - Actual|}{Actual} \times 100$$

Tabular Column:

Actual Distance	Reading 1	Reading 2	Reading 3

Result: